❖ Comptia A+ Assignment Module -1

Module -1, Hardware

1. What is Input Device?

An input device is a device used to give data or instructions to a computer.
Examples: Keyboard, Mouse, Scanner, Microphone

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2. What are Output Devices?

Output devices show the result from the computer.
Examples: Monitor, Printer, Speaker, Projector

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3. What is CPU?

CPU (Central Processing Unit) is the brain of the computer.
It processes data and controls all operations.

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4. What are the types of CPU?

- **Single-core CPU – one core (slow)**
- **Dual-core CPU – two cores**
- **Quad-core CPU – four cores**
- **Octa-core CPU – eight cores (fast)**

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5. What do we need to keep the CPU healthy?

- **Keep it cool (use fan or cooler)**
- **Apply thermal paste**
- **Keep system clean from dust**
- **Use proper power supply**
- **Do not overheat or overload**

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6. Practical: Remove CPU & Apply Thermal Paste

Steps:

1. **Turn off computer and unplug power**
2. **Open CPU cabinet**
3. **Remove CPU cooler carefully**
4. **Take out processor from socket**
5. **Clean old thermal paste**
6. **Apply new thermal paste (small drop)**
7. **Place CPU back and lock it**
8. **Install cooler again**

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7. Practical: Identify CPU & Socket

Steps:

1. **Open computer cabinet**
2. **Find processor on motherboard**
3. **Check CPU name (like Intel i5, AMD Ryzen)**
4. **Identify socket type (like LGA, AM4)**

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8. What is Memory?

Memory stores data and instructions in a computer.

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9. Types of Memory

- **RAM (Random Access Memory) – temporary memory**
- **ROM (Read Only Memory) – permanent memory**

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10. Practical: Identify Memory Types

Steps:

1. **Open system cabinet**
2. **Look at RAM slots**
3. **Check RAM type (DDR3, DDR4, DDR5)**
4. **Check size (4GB, 8GB, etc.)**

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11. Practical: Install Memory (RAM)

Steps:

1. **Turn off computer**
2. **Open cabinet**
3. **Find RAM slot**
4. **Insert RAM properly**
5. **Press until it locks**
6. **Close cabinet and start system**

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12. Practical: Identify Memory Frequency

Steps:

1. **Turn on system**
2. **Open Task Manager**
3. **Go to Performance → Memory**
4. **Check speed (like 2400 MHz, 3200 MHz)**

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13. What is BIOS?

BIOS (Basic Input Output System) is a program stored in motherboard.
It starts the computer and checks hardware.

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14. Working Process of BIOS

1. **When computer starts, BIOS runs**
2. **It checks hardware (RAM, CPU, keyboard)**
3. **Loads operating system (Windows, Linux)**

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15. Practical: Reset BIOS (Soft Reset)

Steps:

1. **Restart computer**
2. **Press key (Delete / F2) to open BIOS**
3. **Select "Load Default Settings"**
4. **Save and exit**

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16. Practical: Hard Reset BIOS

Steps:

1. **Turn off computer**
2. **Remove CMOS battery from motherboard**
3. **Wait 5–10 minutes**
4. **Put battery back**
5. **Start system**

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17. Practical: Identify BIOS Chip

Steps:

1. **Open motherboard**
2. **Find small chip labeled BIOS**
3. **It is near CMOS battery**

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18. What is CMOS?

CMOS is a small memory chip that stores BIOS settings (date, time, boot settings).

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19. What is Motherboard?

Motherboard is the main board of computer.
All parts (CPU, RAM, HDD) connect to it.

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20. Types of Motherboard

- **ATX – large size**
- **Micro-ATX – medium size**
- **Mini-ITX – small size**

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21. Practical: Identify Parts of Motherboard

Steps:

1. **Open cabinet**
2. **Identify parts:**
    - **CPU socket**
    - **RAM slots**
    - **Power connector**
    - **Hard disk connector**
    - **Expansion slots**
3. **Note down all parts**